Type MS · Hexadecimal: why we use it and how to convert

Answer 1 of 7 · Question 3 part a.i · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part a.i in this mark scheme extract

3(a)(i)	Award **one** mark for each correct hexadecimal character in the correct order: • B9		**2**
Question	Answer	Additional Guidance	Mark
3(a)(ii)	**The only correct answer is B** *A is not correct because both are processed at the same speed by a computer* *C is not correct because both use the same amount of storage* *D is not correct because hexadecimal uses number base 16*		**1**
Question	Answer	Additional Guidance	Mark
3(a)(iii)	Award **two** marks for a linked description such as: • (logical/arithmetic) right shift (1) by 2 places (1)		**2**
Question	Answer	Additional Guidance	Mark
3(b)(i)	Award **one** mark for: Pixel/picture element (1)	Accept additional words. Accept incorrect spelling if word is recognisable as pixel	**1**

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Hexadecimal: why we use it and how to convert

Answer 1 of 7 · Question 3 part a.i · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part a.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark
3(b)(ii)	Award **one** mark for each of: • 16x16 pixels (1) • Multiplying (16x16) by 2 bits per pixel in the numerator for colour depth (1) • Dividing by 8 in the denominator to convert to bytes (1) • Dividing 1000 by the number of bytes used for one image (1) Example: $1000 / \frac{16 \times 16 \times 2}{8}$ bytes Allow follow-through	4 marks for any of: 1000 X 8 / 16 x 16 x 2 8000 / 256 x 2 1000 / 16 x 2 x 2 1000 / 16 x 4 8000 / 512	4
Question	Answer	Additional Guidance	Mark
3(b)(iii)	Award up to **two** marks for a linked explanation such as: • high compression ratio (1) repeated/continuous runs (of the same pixel colour) (1) • requires less storage space (1) encoding 16W for line 1 (1)	Allow valid RLE examples from the image. Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
Question	Answer	Additional Guidance	Mark
3(c)	Award **one** mark for each correct cell: Encrypted text / Plain text RED COT	Ignore capitalisation Must be correct spelling	2
		Total for question 3	14

Type MS · Hexadecimal: why we use it and how to convert

Answer 1 of 7 · Question 3 part a.i · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part a.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Hexadecimal: why we use it and how to convert

Answer 2 of 7 · Question 2 part b · 1 mark

2020 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

2(a)(i)	1101 1110 Award one mark for: • MSB = 1 (1) • Rest of pattern correct (1)		2

Question Number	Answer	Additional Guidance	Mark
2(a)(ii)	- (1) 119 (1)		2

Question Number	Answer	Additional Guidance	Mark
2(b)	B 16		1

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	128	Allow the range 0-127	1

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	C		1

Question Number	Answer	Additional Guidance	Mark

8

Type MS · Hexadecimal: why we use it and how to convert

Answer 2 of 7 · Question 2 part b · 1 mark

2020 NOV · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

2(c)(iii)	Award **two** marks for a linked explanation such as: (Before Unicode existed) there are hundreds of different encoding systems (1) and no single encoding system could contain enough characters to represent all major languages (1) Standard ASCII only provides 128 different patterns (1) can't represent all major languages/symbols/characters (1) Unicode uses a minimum of 16 bits (1) so can represent at least 2^{16} characters (1) Unicode has very large number of characters (1) can represent all languages/ASCII was developed (just) for English (1)		2

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Hexadecimal: why we use it and how to convert

Answer 2 of 7 · Question 2 part b · 1 mark

2020 NOV · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Hexadecimal: why we use it and how to convert

Answer 3 of 7 · Question 1 part a · 1 mark

2021 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Number			
1(a)	**The only correct answer is B** ***A*** *is not correct because computers only understand binary* ***C*** *is not correct because Hex values will be held as binary so take up the same space* ***D*** *is not correct because Hex values are held as binary and so take the same time to execute*		1

Question Number	Answer	Additional Guidance	Mark
1(b)	Award **one** mark for each of: • 5 • A	Must be in the correct order	2

Question Number	Answer	Additional Guidance	Mark
1(c)	**The only correct answer is B** ***A*** *is not correct because this is the number of binary patterns that can be represented by 7 bits* ***C*** *is not correct because this is the number of binary patterns that can be represented by 9 bits* ***D*** *is not correct because this is the number of binary patterns that can be represented by 10 bits*		1

Question Number	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for each nibble 0110 0101		2

Question Number	Answer	Additional Guidance	Mark
1(e)	Award **one** mark from: • 01000 (1) • 1000 (1)		1

Type MS · Hexadecimal: why we use it and how to convert

Answer 3 of 7 · Question 1 part a · 1 mark

2021 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(f)	7		1

Question Number	Answer	Additional Guidance	Mark
1(g)(i)	Award **one** mark from: • The smallest block (of colour) in an image (1) • A single dot in a picture (1) • A picture element (1) • A single element of a digital screen (1)		1
1(g)(ii)	Award **one** mark from: • 8 x 9 (1) • 9 x 8 (1)	Accept 72	1
1(g)(iii)	4		1

Question Number	Answer	Additional Guidance	Mark

Type MS · Hexadecimal: why we use it and how to convert

Answer 4 of 7 · Question 1 part b · 2 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Number			
1(a)	**The only correct answer is B** ***A*** *is not correct because computers only understand binary* ***C*** *is not correct because Hex values will be held as binary so take up the same space* ***D*** *is not correct because Hex values are held as binary and so take the same time to execute*		1

Question Number	Answer	Additional Guidance	Mark
1(b)	Award **one** mark for each of: • 5 • A	Must be in the correct order	2

Question Number	Answer	Additional Guidance	Mark
1(c)	**The only correct answer is B** ***A*** *is not correct because this is the number of binary patterns that can be represented by 7 bits* ***C*** *is not correct because this is the number of binary patterns that can be represented by 9 bits* ***D*** *is not correct because this is the number of binary patterns that can be represented by 10 bits*		1

Question Number	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for each nibble 0110 0101		2

Question Number	Answer	Additional Guidance	Mark
1(e)	Award **one** mark from: • 01000 (1) • 1000 (1)		1

Type MS · Hexadecimal: why we use it and how to convert

Answer 4 of 7 · Question 1 part b · 2 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(f)	7		1

Question Number	Answer	Additional Guidance	Mark
1(g)(i)	Award **one** mark from: • The smallest block (of colour) in an image (1) • A single dot in a picture (1) • A picture element (1) • A single element of a digital screen (1)		1
1(g)(ii)	Award **one** mark from: • 8 x 9 (1) • 9 x 8 (1)	Accept 72	1
1(g)(iii)	4		1

Question Number	Answer	Additional Guidance	Mark

Type MS · Hexadecimal: why we use it and how to convert

Answer 5 of 7 · Question 1 part a.iii · 2 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part a.iii in this mark scheme extract

1(ai)	0100 1101 100 1101 Award **one** mark for each correct nibble	Must be in the correct order	2
1(a)(ii)	1000 0010 Award **one** mark for each correct nibble	Must be in the correct order	2
1(a)(iii)	Award **one** mark for each of: • C (1) • 6 (1)	Must be in the correct order	2

Type MS · Hexadecimal: why we use it and how to convert

Answer 5 of 7 · Question 1 part a.iii · 2 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part a.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	400 pixels x 200 pixels x 12 bits / 8 bits per byte Allow expression in word form $\frac{400 \times 200 \times 12}{8}$ 400 x 200 **OR** 80000(1) 12/8 **OR** 1.5 (1)	Accept any other use of 400, 200, 12, 8 that gives the correct answer Allow one mark for 120,000 with no expression	2
1(b)(ii)	Spaces are to help legibility. Award **three** marks for: Line 1 3W 1B 4W Line 2 1W 6B 1W Line 3 1B 2W 1B 1W 1B 2W Line 4 3W 2B 3W Award **two** marks for any three lines correctly encoded Award **one** mark for any two lines correctly encoded	Allow **one** error in the sequence Accept letter before number but must be consistently used	3
1(b)(iii)	Award **one** mark from: • None of the original (image) detail/quality is lost (when the image is stored) (1) • None of the data/information is (permanently) removed (1) • It can be decompressed without losing detail (1)		1

Type MS · Hexadecimal: why we use it and how to convert

Answer 5 of 7 · Question 1 part a.iii · 2 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part a.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	Award up to **three** marks for a linked description: • set the sample rate/parameters/bit-depth (1) • sample (the analogue sound) (1) • measure the sound amplitude/volume/frequency (1) • give a (binary) value/number for each measurement (1) • store data as sample rate and values / digital signals (1)		3
1(c)(ii)	Award **one** mark from: • sound may be missing some frequencies (1) • some audio information/data is lost/deleted during the (compression) process (1)		1
		Total	**16**

Type MS · Hexadecimal: why we use it and how to convert

Answer 5 of 7 · Question 1 part a.iii · 2 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part a.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Hexadecimal: why we use it and how to convert

Answer 6 of 7 · Question 2 part a · 2 marks

2023 JUN · Paper 01 · extract 1 of 5 · find part a in this mark scheme extract

<table>
<tr>
<td>2(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Denary</th><th>Hexadecimal</th></tr>
<tr><td>8</td><td>8 (1)</td></tr>
<tr><td>12</td><td>C (1)</td></tr>
</table>
</td>
<td></td>
<td>2</td>
</tr>
</table>

Type MS · Hexadecimal: why we use it and how to convert

Answer 6 of 7 · Question 2 part a · 2 marks

2023 JUN · Paper 01 · extract 2 of 5 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(b)	**The only correct answer is D** ***A*** *is not correct because* 6^2 *stores 36 patterns, not 64 patterns* ***B*** *is not correct because* **6 × 2** stores 12 patterns, not 64 patterns ***C*** *is not correct because* 6^2 **– 1** stores 35 patterns, not 64 patterns		1

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	Award **one** mark for: • 0001 0111 (1)	Ignore spaces	1

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	Award **one** mark for: • 1111 1000 (1)	Ignore spaces	1

Type MS · Hexadecimal: why we use it and how to convert

Answer 6 of 7 · Question 2 part a · 2 marks

2023 JUN · Paper 01 · extract 3 of 5 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(d)(i)	Award **one** mark for each correct nibble: • 0011 1100	Ignore spaces	2

Question Number	Answer	Additional Guidance	Mark
2(d)(ii)	Award **one** mark for each of: • Negative sign (1) • 22 (1) Example: • -22	Note 22 is calculated from 16 + 4 + 2	2

Question Number	Answer	Additional Guidance	Mark
2(d)(iii)	Award **one** mark for each correct nibble: • 0001 1011	Ignore spaces	2

Type MS · Hexadecimal: why we use it and how to convert

Answer 6 of 7 · Question 2 part a · 2 marks

2023 JUN · Paper 01 · extract 4 of 5 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(e)	Award **one** mark for each of: 1. 13 x 1000 // 13000 in the numerator (1) 2. 1024 in the denominator (1) Examples: $\frac{13 \times 1000}{1024}$ $\frac{13000}{1024}$ $\frac{13 \times 500}{512}$ $\frac{13 \times 250}{256}$ $\frac{13 \times 125}{128}$ $\frac{13 \times 10^3}{2^{10}}$	Award reduced expressions, if correct Allow all equivalent exponents Note 1000/1024 = 1.024, so could have 13 * 1.024	2

Type MS · Hexadecimal: why we use it and how to convert

Answer 6 of 7 · Question 2 part a · 2 marks

2023 JUN · Paper 01 · extract 5 of 5 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Hexadecimal: why we use it and how to convert

Answer 7 of 7 · Question 3 part a.ii · 1 mark

2025 JUN · Paper 01 · extract 1 of 3 · find part a.ii in this mark scheme extract

3(a)(i)	Award **one** mark for each correct hexadecimal character in the correct order: • B9		2
Question	Answer	Additional Guidance	Mark
3(a)(ii)	**The only correct answer is B** *A is not correct because both are processed at the same speed by a computer* *C is not correct because both use the same amount of storage* *D is not correct because hexadecimal uses number base 16*		1
Question	Answer	Additional Guidance	Mark
3(a)(iii)	Award **two** marks for a linked description such as: • (logical/arithmetic) right shift (1) by 2 places (1)		2
Question	Answer	Additional Guidance	Mark
3(b)(i)	Award **one** mark for: Pixel/picture element (1)	Accept additional words. Accept incorrect spelling if word is recognisable as pixel	1

Type MS · Hexadecimal: why we use it and how to convert

Answer 7 of 7 · Question 3 part a.ii · 1 mark

2025 JUN · Paper 01 · extract 2 of 3 · find part a.ii in this mark scheme extract

<table>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)(ii)</td><td>Award one mark for each of:

• 16x16 pixels (1)
• Multiplying (16x16) by 2 bits per pixel in the numerator for colour depth (1)
• Dividing by 8 in the denominator to convert to bytes (1)
• Dividing 1000 by the number of bytes used for one image (1)

Example:

$1000 / \frac{16 \times 16 \times 2}{8}$ bytes

Allow follow-through</td><td>4 marks for any of:
1000 X 8 / 16 x 16 x 2
8000 / 256 x 2
1000 / 16 x 2 x 2
1000 / 16 x 4
8000 / 512</td><td>4</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)(iii)</td><td>Award up to two marks for a linked explanation such as:

• high compression ratio (1) repeated/continuous runs (of the same pixel colour) (1)
• requires less storage space (1) encoding 16W for line 1 (1)</td><td>Allow valid RLE examples from the image.

Explanation can be a combination of different marking points. E.g. MP1a and MP2b</td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(c)</td><td>Award one mark for each correct cell:

<table><tr><th>Encrypted text</th><th>Plain text</th></tr><tr><td>  </td><td>RED</td></tr><tr><td>  </td><td>COT</td></tr></table></td><td>Ignore capitalisation
Must be correct spelling</td><td>2</td></tr>
<tr><td colspan="3">Total for question 3</td><td>14</td></tr>
</table>

10

Type MS · Hexadecimal: why we use it and how to convert

Answer 7 of 7 · Question 3 part a.ii · 1 mark

2025 JUN · Paper 01 · extract 3 of 3 · find part a.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark